

Giridhari Pal

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PROFESSIONAL SUMMARY

Backend engineer with 3+ years of production experience designing and maintaining scalable backend services and APIs. Strong expertise in RESTful API development, microservices architecture, asynchronous processing, and event-driven systems using Python, Rust, FastAPI, Kafka, and PostgreSQL. Experienced in building secure, cloud-deployed backend platforms supporting real-time analytics and large-scale data pipelines. Hands-on experience with SQL, NoSQL, and data pipelines supporting AI model integration and inference workflows.

TECHNICAL SKILLS

Languages: Python, Rust

Backend & API: FastAPI, Actix-Web, Node.js, Django, gRPC

Architecture: Microservices, Event-Driven Architecture, Distributed Systems, Async Processing

Databases: PostgreSQL, MongoDB, InfluxDB, Redis, DuckDB, SQLite

IoT & Streaming: ESP32, OTA, Kafka, MQTT

Cloud & Infrastructure: AWS (EC2, S3), DigitalOcean, Utho Cloud, Docker, Linux (Ubuntu Server)

Frontend: ReactJS, TypeScript, Tauri, PyQt5

PROFESSIONAL EXPERIENCE

Dhirubhai Ambani University

Aug 2025 – Present

Junior Research Fellow — Backend & Data Systems

Gujarat, India

- Led and owned the end-to-end system architecture for an Energy Management System integrating ESP32-based IoT devices with backend services, addressing the challenge of reliable, continuous telemetry ingestion from distributed hardware.
- Architected real-time data pipelines using MQTT and Kafka, supporting high-throughput telemetry for uninterrupted operation.
- Introduced a secure Actix-Web-based OTA update service to manage firmware versioning, device compatibility validation, and controlled update distribution, improving device reliability and operational safety.
- Deployed containerized backend microservices using Docker, MQTT, Kafka, gRPC, Actix-Web, PostgreSQL, InfluxDB, and Redis to support scalable, fault-tolerant processing under sustained data loads. .

Independent Technical

Oct 2024 – Aug 2025

Consultant / Freelance Engineer

Remote

- **Full-Stack & DevOps:** Architected custom ERP and Leave Management systems using FastAPI, React, and PostgreSQL handled deployment via Docker on cloud infrastructure.
- **Systems Integration:** Integrated ZWCAD with FastAPI backends using VBA and Tauri to automate industrial design-to-inventory workflows for manufacturing clients.
- **Embedded Systems:** Programmed ESP32 controllers for solar tracking, focusing on sensor fusion and stable motor control.
- **Machine Learning:** Leveraged OpenCV to build an OCR engine for banking document automation, significantly reducing manual data entry errors.

RC Labs — Battery Management System (BMS)

Nov 2021 – Oct 2024

Intern → SDE & DevOps Consultant

Bangalore, India

- Owned the end-to-end backend and data visualization platform integrating IoT telemetry, data pipelines, and monitoring systems across desktop, web, and cloud environments.
- Architected MQTT and Kafka real-time data pipelines to process millions of data points daily with sub-100ms latency, and gRPC, PostgreSQL, and Grafana to support continuous, high-frequency telemetry ingestion and long-running workloads.
- Established a secure and scalable system architecture incorporating custom encryption, multithreading, multiprocessing, and containerized services, significantly improving throughput, system stability, and fault tolerance.
- Enabled operational simplicity by standardizing deployments using Docker and centralized dashboards, allowing reliable monitoring, historical analysis, and remote access across production environments.

EDUCATION

Sri Krishna Institute of Technology (VTU)

2019 – 2023

B.E. in Information Science and Engineering (CGPA: 7.5)

Bangalore, India